

Experiment 6

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(A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

SOLUTION-

SELECT

sale_date,

SUM(CASE WHEN fruit = 'apples' THEN sold_num ELSE -sold_num END)

AS diff

FROM Sales**GROUP BY** sale_date**ORDER BY** sale_date;

(B)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

id represents the current node.

p_id represents the parent node.

Classify every node into one of the following categories:

Root: A node that does not have any parent.

Leaf: A node that has a parent but does not have any children.

Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tree

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Expected Output

id	Type
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

SOLUTION-

```
SELECT
  id,
  CASE
    WHEN p_id IS NULL THEN 'Root'
    WHEN id IN (SELECT DISTINCT p_id
                FROM Tree
                WHERE p_id IS NOT NULL) THEN 'Inner'
    ELSE 'Leaf'
  END AS Type
FROM Tree
ORDER BY id;
```